

## Inputs (Sec. 4)

TMYx weather: Riyadh, Jeddah  
floor plan: 132 zones, 60 units,  
9 meters  
SBC 602 and pre-code envelopes  
unit catalogue; SASO 2663 COP  
two-tier volume tariff  
comfort band [22, 24] °C

## 1 Room-resolved model (Sec. 5)

two-node rooms coupled through  
walls, slabs and open doors  
heat pumps: thermostat, modes,  
restart delay, cycling losses  
latent load; per-meter bill

## 2 Formal problem (Sec. 5)

multi-mode system: timed actions,  
leaps, infinite schedules,  
limit-average cost  
minimise the annual bill  
s.t. conditioned rooms  $\leq 24$  °C  
policy: setpoints, pre-cooling

## 3 Theory (Sec. 6)

optimal leap: highest level,  
lowest feasible mode  
infinite-horizon optimum  
pre-cooling bound  
sources of saving  
→ conditions to check

## 4 Optimisation, evaluation (Sec. 7)

optimal control (LP) of the  
idealised model  
Alg. 1: annual simulation,  
per-meter bill  
Alg. 2: 49 thermostat settings  
select on 60 days, evaluate on 365

## 5 Verification & validation (Sec. 8)

time-step convergence  
heat balance of passive rooms  
BESTEST-style 600/900/640/940  
EnergyPlus on the same storeys,  
with and without setbacks  
stock energy use (EUI)

## 6 Experiments (Sec. 9)

one lever at a time: setpoint,  
mode, envelope, equipment,  
scheduling  
3 cases  $\times$  2 cities  $\times$  2 seasons  
sensitivity analysis

## Outputs (Secs. 9–11)

saving of each lever by case,  
city and season  
SAR a year per apartment  
comfort in rooms without units  
which lever pays under a  
volume tariff

policy  $\theta$  and constraints (C1)–(C4)

conditions checked